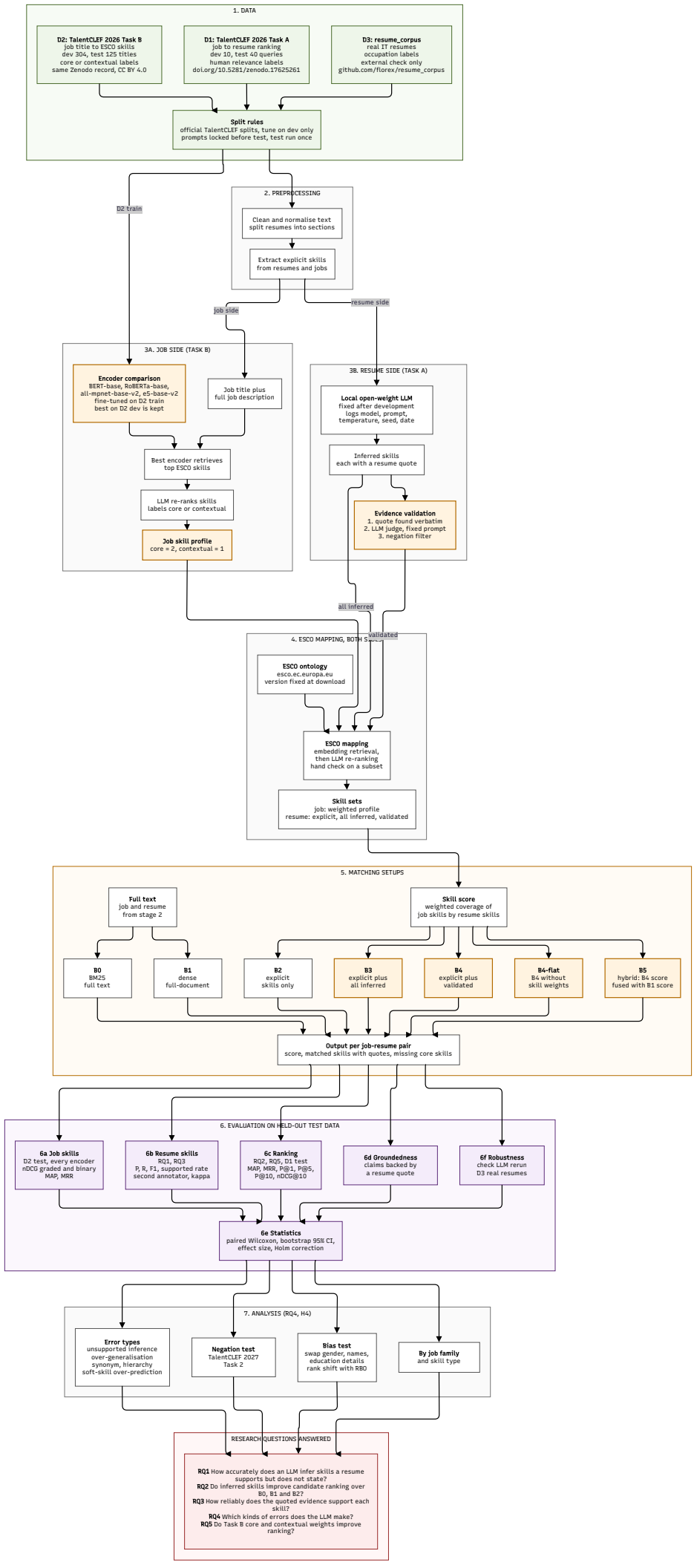

1. DATA
D2: TalentCLEF 2026 Task B
job title to ESCO skills
dev 304, test 125 titles
core or contextual labels
same Zenodo record, CC BY 4.0
D1: TalentCLEF 2026 Task A
job to resume ranking
dev 10, test 40 queries
human relevance labels
doi.org/10.5281/zenodo.17625261
D3: resume_corpus
real IT resumes
occupation labels
external check only
github.com/florex/resume_corpus
Split rules
official TalentCLEF splits, tune on dev only
prompts locked before test, test run once
D2 train
2. PREPROCESSING
Clean and normalise text
split resumes into sections
Extract explicit skills
from resumes and jobs
job side
resume side
3A. JOB SIDE (TASK B)
Encoder comparison
BERT-base, RoBERTa-base,
all-mpnet-base-v2, e5-base-v2
fine-tuned on D2 train
best on D2 dev is kept
Job title plus
full job description
Best encoder retrieves
top ESCO skills
LLM re-ranks skills
labels core or contextual
Job skill profile
core = 2, contextual = 1
3B. RESUME SIDE (TASK A)
Local open-weight LLM
fixed after development
logs model, prompt,
temperature, seed, date
Inferred skills
each with a resume quote
Evidence validation
1. quote found verbatim
2. LLM judge, fixed prompt
3. negation filter
all inferred
validated
4. ESCO MAPPING, BOTH SIDES
ESCO ontology
esco.ec.europa.eu
version fixed at download
ESCO mapping
embedding retrieval,
then LLM re-ranking
hand check on a subset
Skill sets
job: weighted profile
resume: explicit, all inferred, validated
5. MATCHING SETUPS
Full text
job and resume
from stage 2
Skill score
weighted coverage of
job skills by resume skills
B0
BM25
full text
B1
dense
full-document
B2
explicit
skills only
B3
explicit plus
all inferred
B4
explicit plus
validated
B4-flat
B4 without
skill weights
B5
hybrid: B4 score
fused with B1 score
Output per job-resume pair
score, matched skills with quotes, missing core skills
6. EVALUATION ON HELD-OUT TEST DATA
6a Job skills
D2 test, every encoder
nDCG graded and binary
MAP, MRR
6b Resume skills
RQ1, RQ3
P, R, F1, supported rate
second annotator, kappa
6c Ranking
RQ2, RQ5, D1 test
MAP, MRR, P@1, P@5,
P@10, nDCG@10
6d Groundedness
claims backed by
a resume quote
6f Robustness
check LLM rerun
D3 real resumes
6e Statistics
paired Wilcoxon, bootstrap 95% CI,
effect size, Holm correction
7. ANALYSIS (RQ4, H4)
Error types
unsupported inference
over-generalisation
synonym, hierarchy
soft-skill over-prediction
Negation test
TalentCLEF 2027
Task 2
Bias test
swap gender, names,
education details
rank shift with RBO
By job family
and skill type
RESEARCH QUESTIONS ANSWERED
RQ1 How accurately does an LLM infer skills a resume supports but does not state?
RQ2 Do inferred skills improve candidate ranking over B0, B1 and B2?
RQ3 How reliably does the quoted evidence support each skill?
RQ4 Which kinds of errors does the LLM make?
RQ5 Do Task B core and contextual weights improve ranking?